BANK CUSTOMER DATA ANALYSIS USING CORRELATION AND COVARIANCE MEASURES

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1. INTRODUCTION

PROBLEM

- Identify customer patterns?
- How are age, balance activity related?
- Which variables show strong relationships with each other?

APPROACH

- Collect data using a questionnaire.
- Clean and organize the collected data.
- Perform statistical analysis.
- Visualize the results.
- Interpret the findings.

WHY THIS STUDY?

- Understand the behaviour/pattern.
- Identify important factors.
- Study relationships between variables.
- Support banking decisions.

2. DATASET

SOURCE:

Bank Customer Dataset

TOTAL RESPONSES:

10,000 Customers

KEY VARIABLES COLLECTED

- Customer ID, Credit Score
- Geography, Gender
- Age& Tenure
- Balance, Number of Products
- Credit Card Status
- Active Member Status
- Estimated Salary
- Customer Status

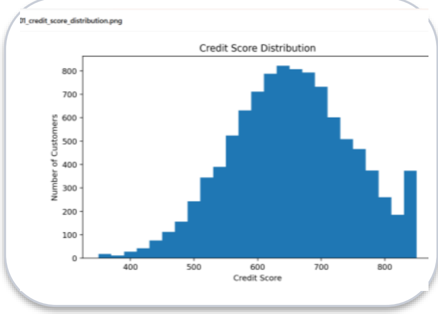
DATA

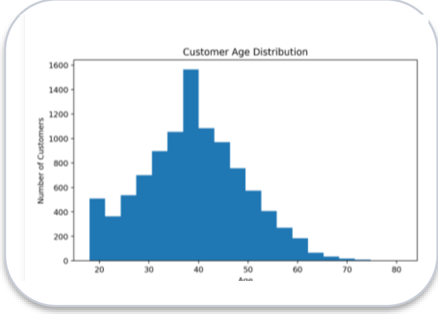
10,000 Records
14 Variables
Numerical &Data

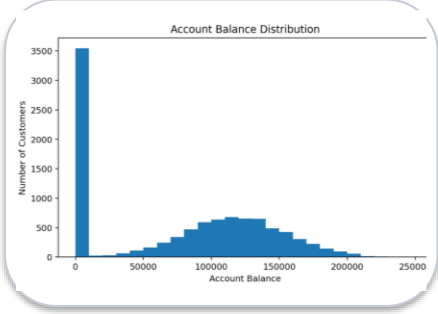
3. METHOD PIPELINE

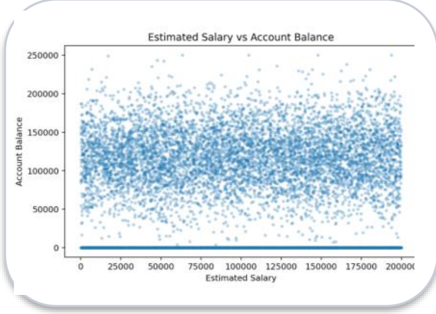
DATA COLLECTION
↓
DATA CLEANING
↓
DESCRIPTIVE STATISTICS
↓
EXPLORATORY DATA ANALYSIS
↓
CORRELATION ANALYSIS
↓
HYPOTHESIS TESTING
↓
VISUALIZATION & INTERPRETATION

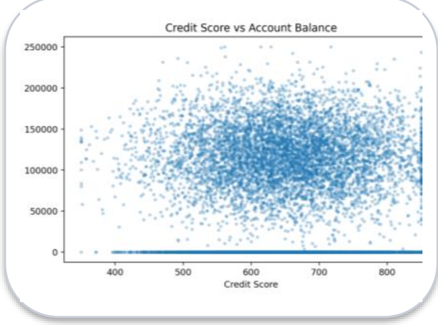
4. RESULTS

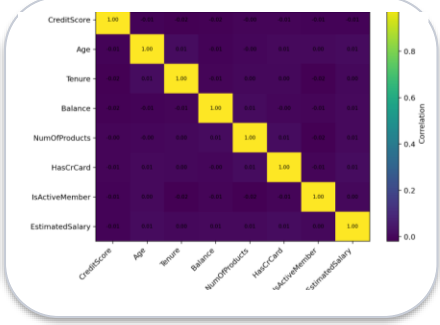


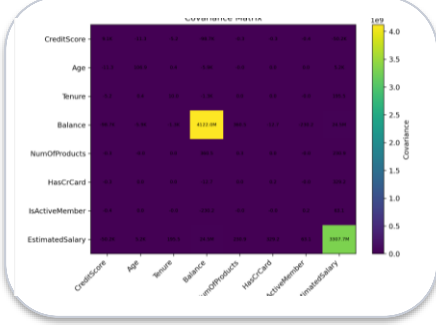


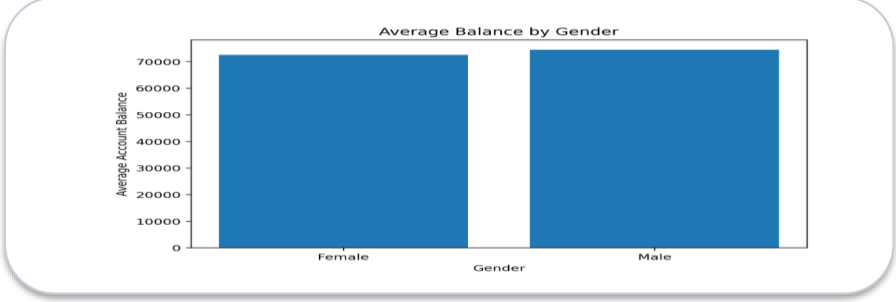


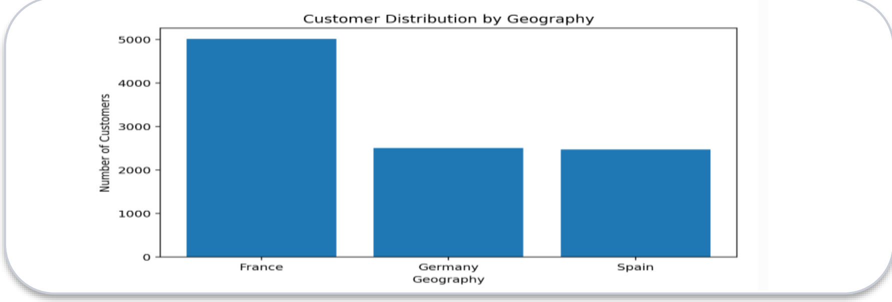












5. CONCLUSION

- Correlation identifies relationships between customer variables.
- Covariance shows how financial variables change together.
- Customer age, salary and balance show useful patterns.

6. FUTURE DIRECTIONS

- Expand the study using larger customer datasets.
- Include transaction and spending behaviour.
- Develop predictive models for customer behaviour.

KEY INSIGHT

- 10,000 Customers | 14 Variables
- Correlation identifies relationships.
- Covariance measures joint changes.
- Analysis supports better decisions.